Lahars

What are they, what can they do, and what do we need to know to model them?



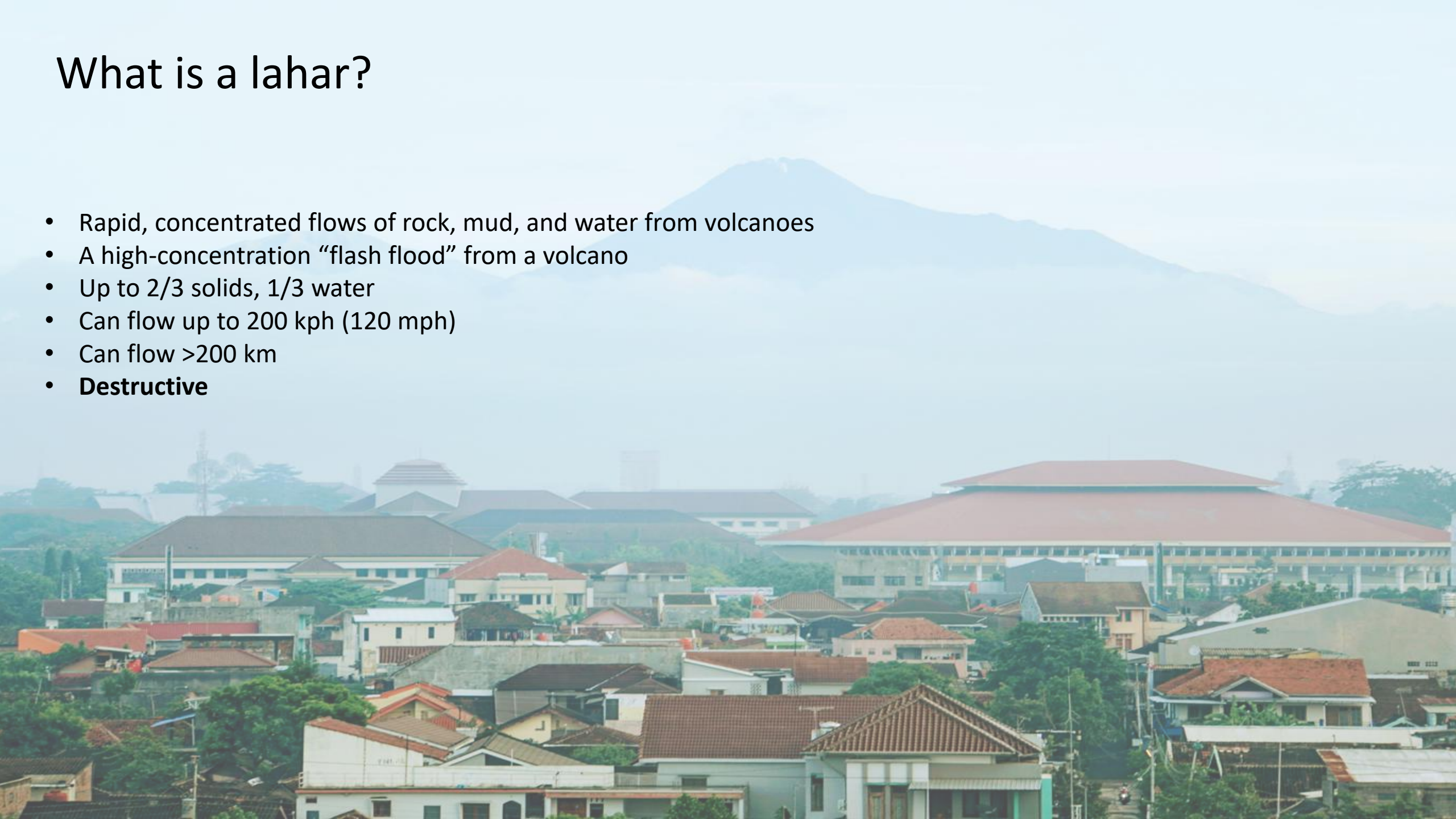


VDAP

USGS

What is a lahar?

- Rapid, concentrated flows of rock, mud, and water from volcanoes
- A high-concentration “flash flood” from a volcano
- Up to 2/3 solids, 1/3 water
- Can flow up to 200 kph (120 mph)
- Can flow >200 km
- **Destructive**



Lahars can occur

WITH or WITHOUT eruptions!!

Triggered by:

- Rapidly melting snow/ice (from dome growth, PDCs, incandescent ejecta, lava flows)
- Heavy rain on tephra or other sediment (meteoric or eruption-induced)
- Lake breakouts (eruption-related or not)
- Sediment dam breakouts (eruption-related or not)
- Glacier outburst floods (jökulhlaups caused by eruptions under glaciers)
- Gravitational collapse of edifice (landslides)
- Groundwater expulsion







VolFilm series: Lahars – A River of Rock https://www.youtube.com/watch?v=ASu2yW9mb64&list=PLfQrRLVgrlwjIEeiAivbQ-Kl39E5RbdiX&index=6

August 1986: VDAP was formed by official agreement between USGS and USAID:

Better to prepare than repair

- Mission: VDAP's mission is to reduce loss of life and property and limit the economic impact of foreign volcanic crises, thereby preventing such crises from becoming disasters.
- Initially a team of 5 people and a budget of ~400K$ (~1.1M$)
- Originally focused on the Latin American and Caribbean region
- Now we have a team of ~22 FTEs and a budget of ~8.5M$ with a global mandate
- In 2025 we were moved to the State Department

https://www.youtube.com/@volfilm5670



Flow Types

wet

Stream Flow

Hyperconcentrated Flow

Debris Flow & Lahar

Granular Flow

dry









Volume fraction between 100% fluid flow --- 100% granular flow

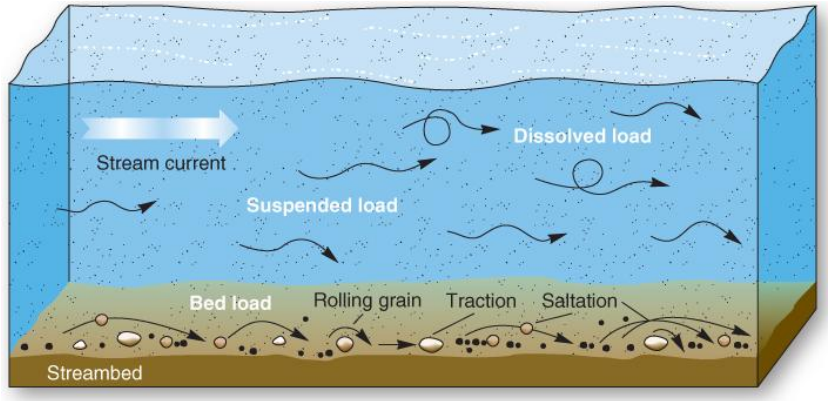
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Flow Types



Streamflow:
Water flow with low suspended sediment load; transport of grains during high flow events by suspension, saltation & bedload dragging; no shear strength; good fishing!



▲Figure 15.13 **Fluvial transport.** Fluvial transportation of eroded materials as dissolved load, suspended load, and bed load. Bed load includes grains transported by traction (rolling or dragging of materials along the streambed) and saltation.

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Hyperconcentrated Flow:
Muddy flow with high suspended sediment load; transport of grains during high flow events by suspension, saltation & bedload dragging;
no matrix strength, not a cohesive flow; erosion and undermining of structures can be very damaging as can the extreme aggradation result in massive, damaging sediment deposits.





Flow Types



Debris Flow and Lahar:
Slurry flow of concrete-like mud and rock with high cohesion where the process is driven by intergranular pressures (pore fluid pressure); has a yield strength to buoyantly transport larger clasts, boulders, trees, bridges; great destruction and loss of life is possible

Granular Flow:
Granular flow or rock avalanche is dominated by frictional flow and collisions



Note: PDCs are considered granular flows, but they are also "fluidized" by gases and particle motion

VDAP

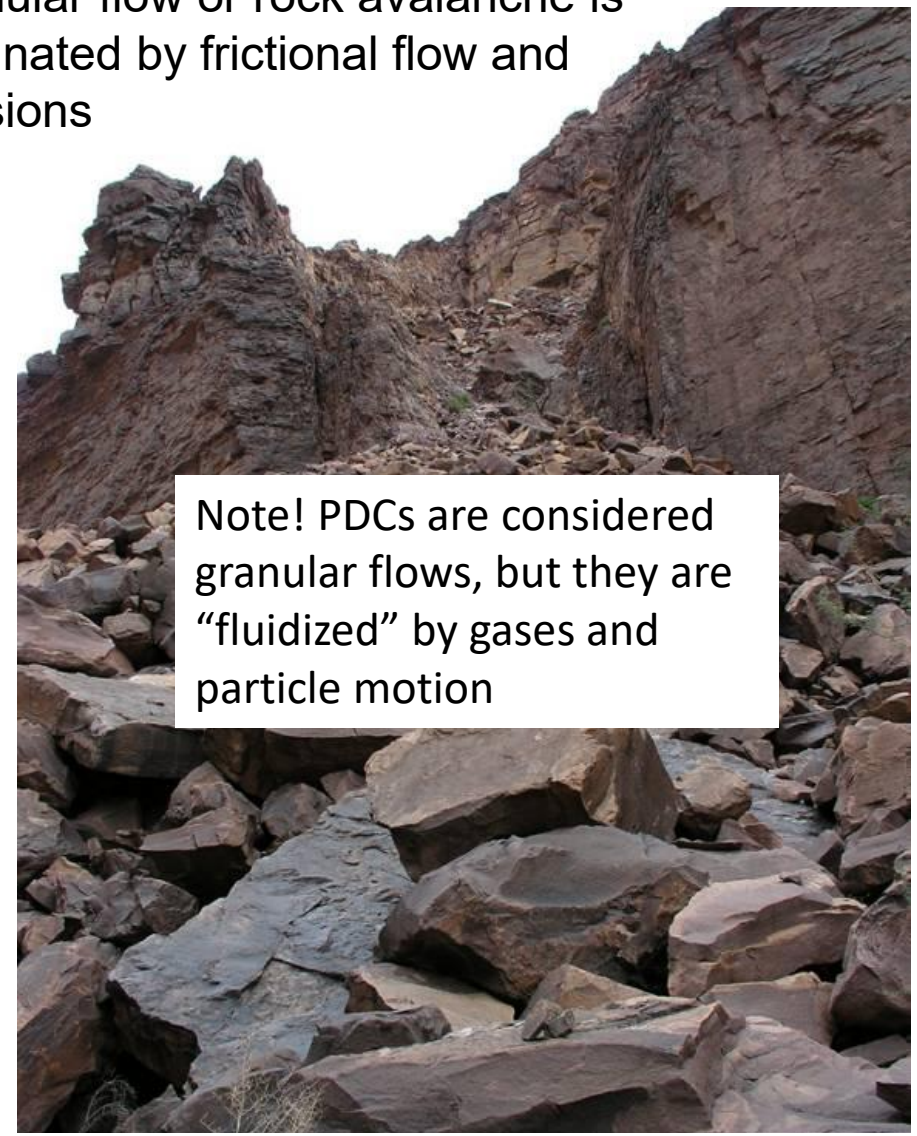
USGS

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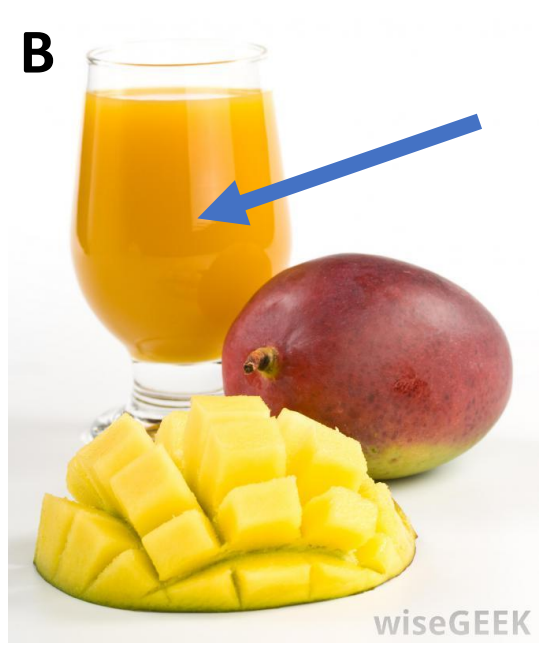
Hyperconcentrated Flow

Debris Flow/Lahar

Granular Flow

Flow Type









Semeru, Indonesia
2003
Franck Levigne

Progression of the 19 January 2002 lahar front in the Curah Lengkong river, Semeru volcano, Indonesia. Franck Lavigne.

https://imaggeo.egu.eu/view/3895/



Fuego, Guatemala

https://youtu.be/5NDfhQY67tI?si=NsF36rZ5EwTxGIJO

Rudiger Escobar Wolf, 2009



VDAP

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Hindu Kush, Afghanistan, 2007 https://www.youtube.com/watch?v=51C7vEAVbxk

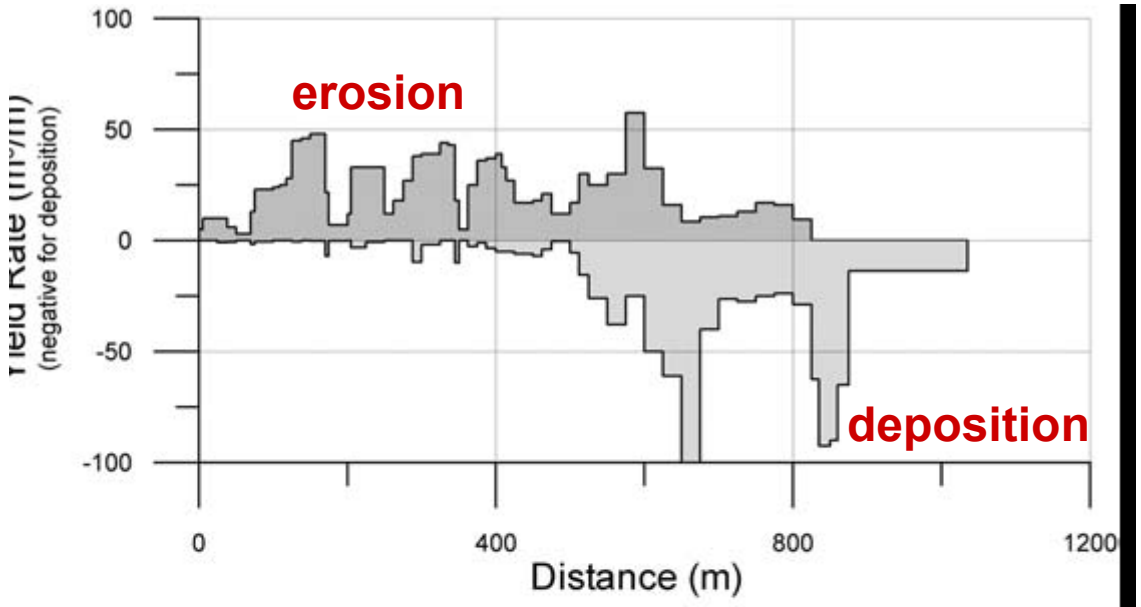
What's happening as the lahar flows?

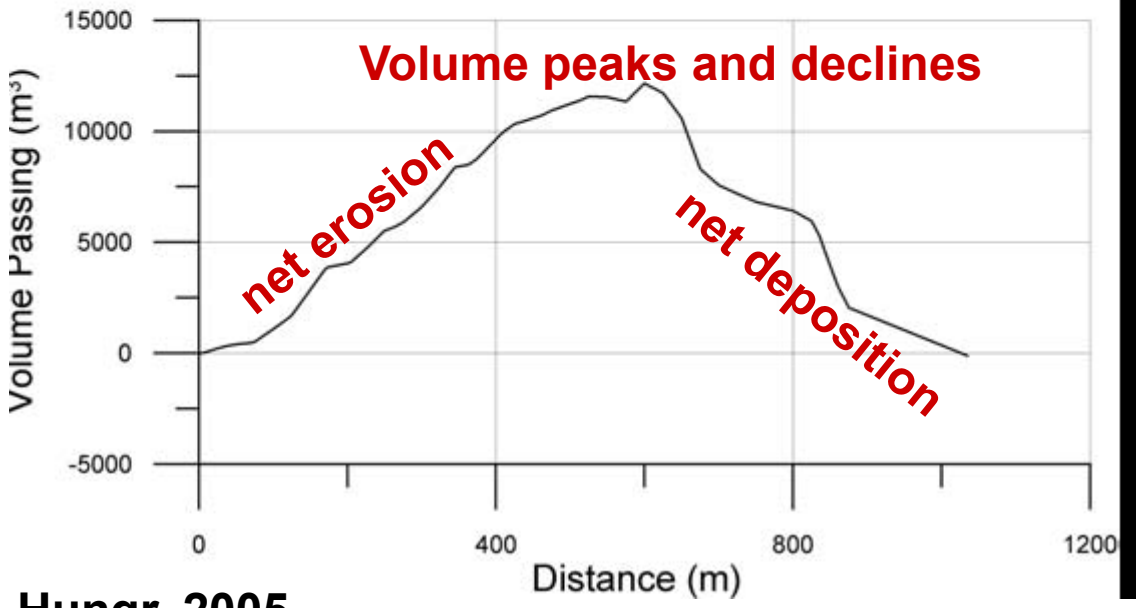
Yield Rate ($\Delta m^3/m$)

Change in volume per meter distance along channel

Flow Volume (m^3)

Mass Balance Analysis

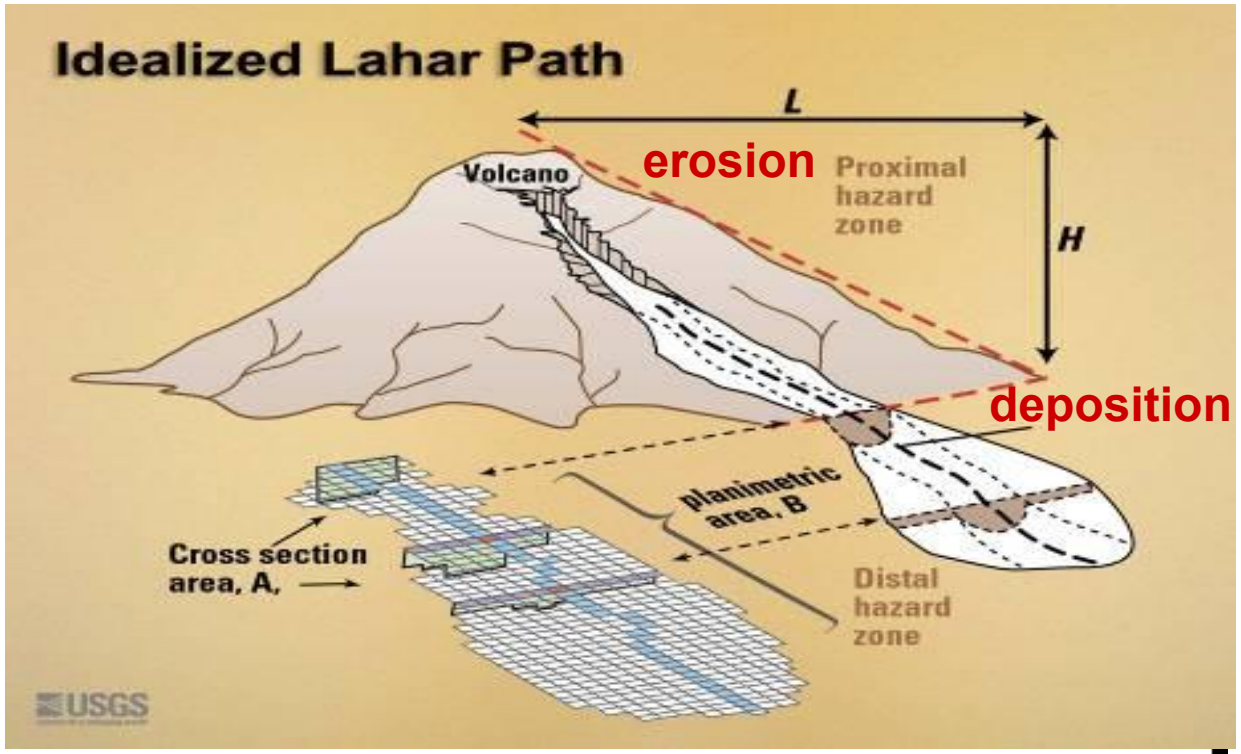


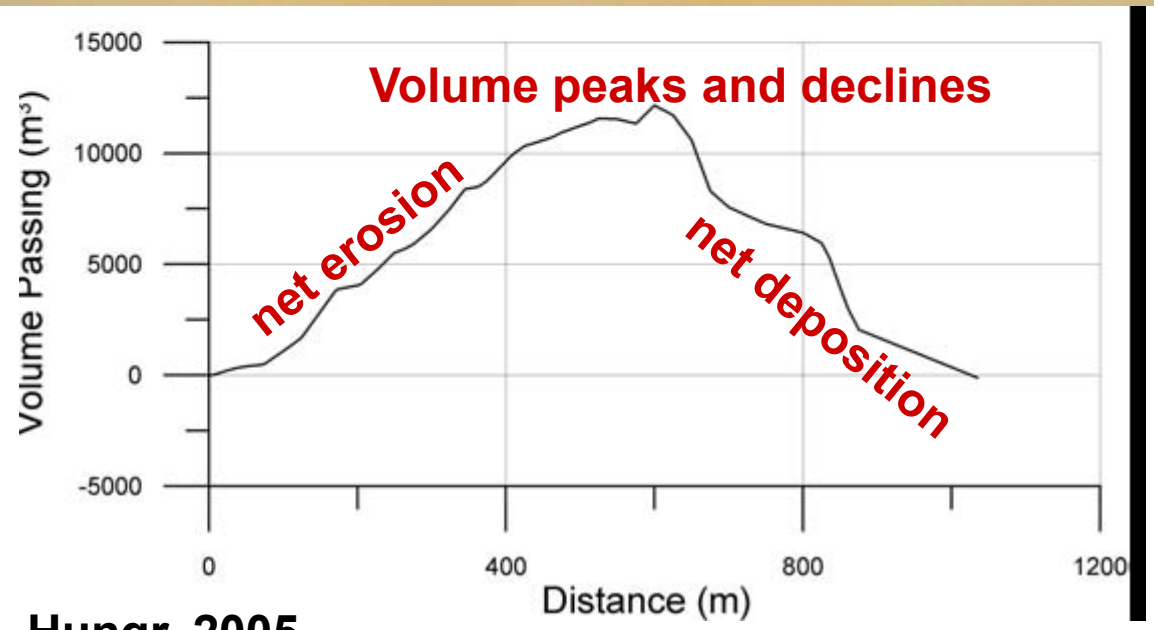


Hungr, 2005

- Mass balance is not constant as flow moves from start to finish; lahar gains volume near the volcano through erosion and lose volume further away through deposition.
- Erosion is greatest on steep slopes where there is much loose volcanic material
- Deposition is greatest on gentle slopes
- Maximum volume along the path marks an important change in lahar process; erosion rate slows and deposition rate increases. This will be important for modeling!

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Hungr, 2005

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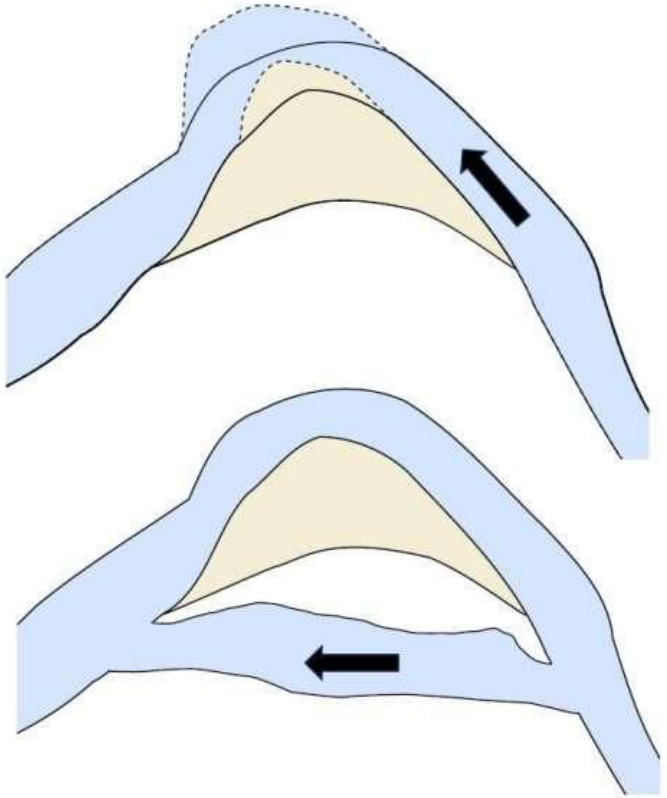


Soufrière Hills Volcano, Montserrat. Sarah Ogburn.

- Long-term hazard
- Cause changes to channels
- Progressively in-fill valleys and effect the runout of other hazards like PDCs

Types of channel migration

- **Lateral Migration**: Gradual movement of river's path across the floodplain.
- **Avulsion**: A river suddenly changes its path.





Pinatubo, Philippines

Rio Paez lahars at Belalcazar, Colombia; 1994, 2007, 2008





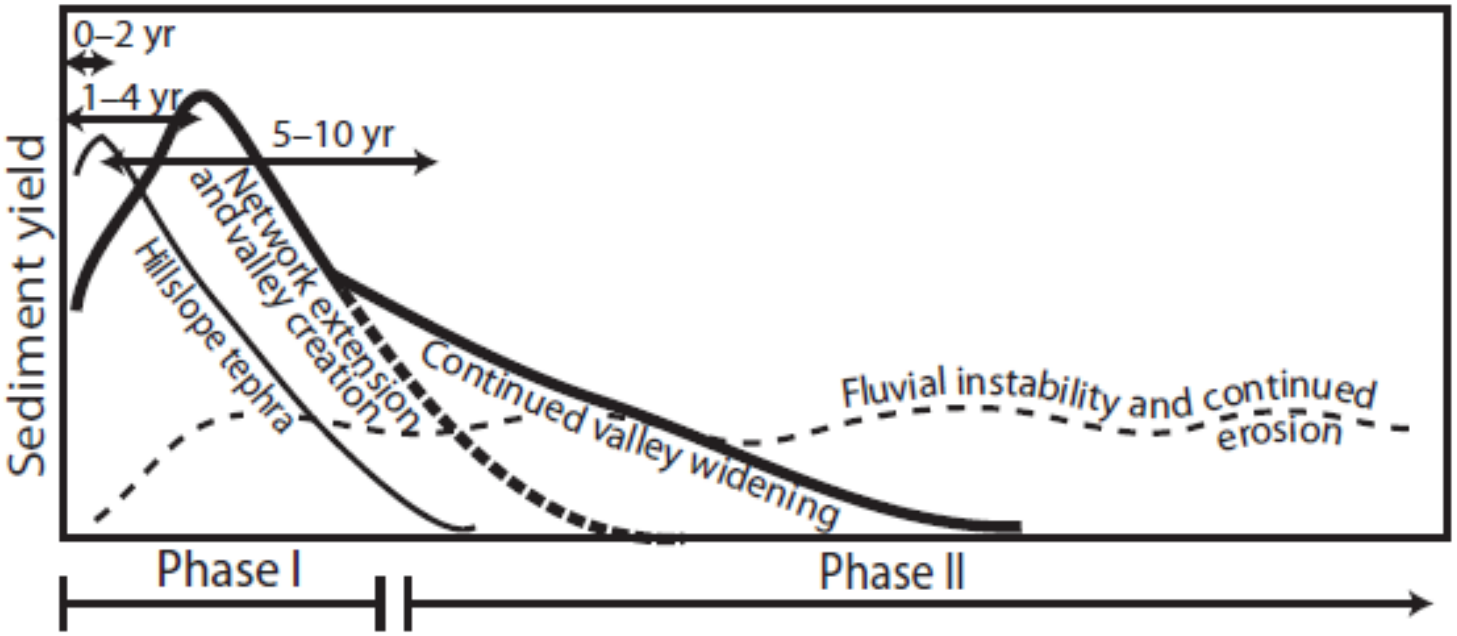




After events, channel may change course or may become choked with sediment that will result in subsequently higher and wider flooding zones; also after some quiescence, dramatic incision may also occur. Updating DEMs will be needed for future hazard maps.

Sediment yields

- Work at Pinatubo and Mt. St. Helens has shown that sediment yields can remain elevated for decades
- Can increase flood peaks and provide source material for lahars
- Schematic illustrating change of dominant sediment sources through time.
- Phase I: hillslope tephra and valley creation lead to high sediment yields early that decline exponentially through time.
- Phase II: continued valley widening and fluvial instability maintain high sediment yields for decades.



Gran et al. 2011

Lahar Deposits
(different from the lahar itself)
Mount St. Helens





Deposit Volume < Flow Volume

Lahar Deposits- Mount St. Helens

